

Free-body diagrams

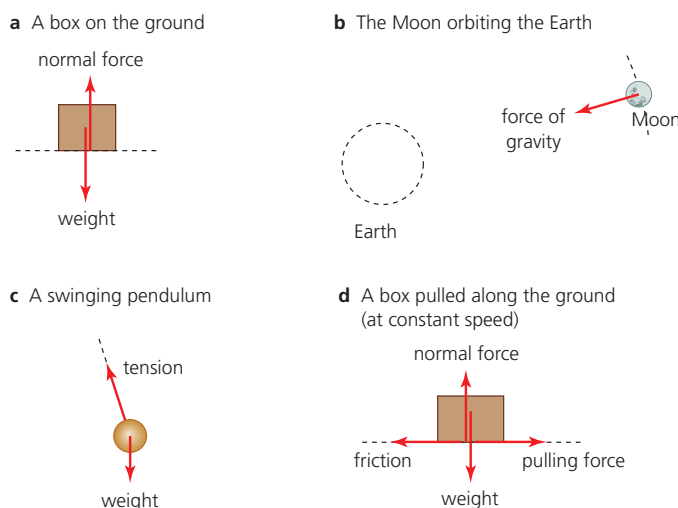
SYLLABUS CONTENT

- Forces acting on a body can be represented in a free-body diagram.

Even the simplest of force diagrams can get confusing if all the forces are shown. To make the diagrams simpler we usually draw only one object and show only the forces acting on that one object. These drawings are called **free-body diagrams**. (Physicists use the words ‘body’ and ‘object’ interchangeably.) Some simple examples are shown in Figure A2.35.

◆ Free-body diagram

Diagram showing all the forces acting on a single object, and no other forces.



■ **Figure A2.35** Free-body diagrams; the object has a solid outline and the forces are shown in red

◆ Point particle, mass or charge

Theoretical concept used to simplify the discussion of forces acting on objects (especially in gravitational and electric fields).

The diagrams are often further simplified by representing the object as a small square, or circle, and considering it to be a **point particle / mass**.

● Nature of science: Models

Point objects, particles and masses

A point particle is an idealized, simplified representation of any object, whatever its actual size and shape. As the name suggests, a point particle does not have any dimensions, or occupy any space. Typically, the ‘point’ will be located at the centre of mass of the object.

When the concept is used, we do not need to consider the complications and variations that are involved with extended objects. For example, if we consider an object as a point particle, all forces act through the same point and analysis can ignore any possible rotational effects caused by the forces acting on it.

Resultant forces and components

SYLLABUS CONTENT

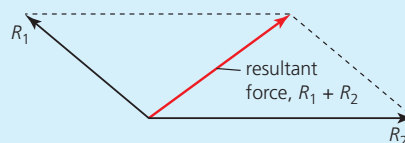
- Free-body diagrams can be analysed to find the resultant force on a system.

Tool 3: Mathematics

Add and subtract vectors in the same plane

Vector addition is an important mathematical skill that occurs in several places in the IB Physics course, but the addition of forces is the most common application. Figure A2.36 shows an example of how to find the resultant of two force vectors.

A **resultant force** is represented in size and direction by the diagonal of the parallelogram (or rectangle) which has the two original force vectors as adjacent sides.



■ **Figure A2.36** Adding two forces to determine a resultant

◆ **Resultant force** The vector sum of the forces acting on an object, sometimes called the unbalanced or net force.

◆ **Resultant** The single vector that has the same effect as the combination of two or more separate vectors.

◆ **Components (of a vector)** Any single vector can be considered as having the same effect as two parts (components) perpendicular to each other.

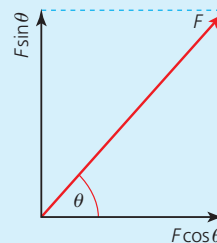
◆ **Inclined plane** Flat surface at an angle to the horizontal (but not perpendicular). A simple device that can be used to reduce the force needed to raise a load; sometimes called a ramp.

Tool 3: Mathematics

Resolve vectors

As we have seen, two forces can be combined to determine a single **resultant**. The 'opposite' process is very useful: a single force, F , can be considered as being equivalent to two smaller forces at right angles to each other. The two separate forces are called **components**.

This process is called resolving a force into two components. It can be used when the original force is not acting in a direction which is convenient for analysis. Because the two components are perpendicular to each other their effects can be considered separately. Figure A2.37 shows how a force can be resolved into two perpendicular components.



■ **Figure A2.37** Force, F , resolved into two components

Any force, F , can be resolved into two independent components which are perpendicular to each other:

$$F \sin \theta \text{ and } F \cos \theta$$

WORKED EXAMPLE A2.6

- a Draw a free-body diagram for an object which is stationary on a slope (**inclined plane**) which makes an angle of 35° with the horizontal.
- b The object has a mass of 12.7 kg and just begins to slide down the slope if the angle is 35° . Using $g = 9.81\text{ N kg}^{-1}$, calculate the component of the weight for this angle:
 - i down the slope
 - ii perpendicular into the slope.
- c State values of the frictional force and the normal force acting on the object.
- d Determine the coefficient of static friction in this situation.

Answer

- a See Figure A2.38, which represents the object as a point. The resultant contact force from the slope on the object must be equal and opposite to the weight, F_g . The contact force can be considered as the combination of two perpendicular components: F_N perpendicular to the slope, and F_f the frictional force stopping the object from sliding down the slope.

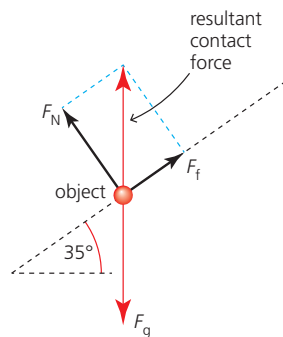


Figure A2.38 Free-body diagram for an object on a slope

Sometimes it is preferred to represent the object as more than just a point. See Figure A2.39 for an example. However, this may cause confusion about exactly where the forces act.

- b See Figure A2.39.

Component down slope $mg \sin 35^\circ = 12.7 \times 9.81 \times 0.574 = 71.5\text{ N}$

Component into slope $= mg \cos 35^\circ = 12.7 \times 9.81 \times 0.819 = 102\text{ N}$

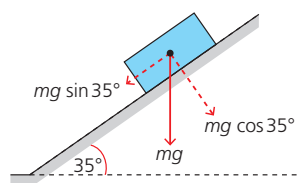


Figure A2.39 Components of weight

- c Frictional force equals component down the slope, but in the opposite direction $= 71.5\text{ N}$ up the slope.
Normal force equals component into the slope, but in the opposite direction $= 102\text{ N}$ upwards.

$$d \quad F_f = \mu_s F_N$$

$$\mu_s = \frac{F_f}{F_N} = \frac{71.5}{102} = 0.70 \text{ (which is equal to } \tan \theta \text{)}$$

- 25 Draw fully labelled free-body diagrams for:

- a a car moving horizontally with a constant velocity
- b an aircraft moving horizontally at constant velocity
- c a boat decelerating after the engine has been switched off
- d a car accelerating up a hill.

- 26 A wooden block of mass 2.7 kg rests on a slope which is inclined at 22° to the horizontal.

- a Make calculations which will enable you to draw a free-body diagram, similar to Figure A2.38, but giving numerical values for the forces.
- b If the angle is increased, the block will slide down the slope. Calculate the coefficient of friction.

- c State whether your answer to part **b** is for static or kinetic friction.

- 27 A pendulum on the end of a string has a mass of 158 g .

- a Draw a free-body diagram representing the situation when the string is making an angle of 20° to the vertical.
- b By adding components of weight to your diagram, show that the tension in the string is 1.5 N .
- c What effect does the other component ($mg \sin \theta$) have on the pendulum?
- d Discuss how the tension in the string changes while the pendulum is swinging from side to side.